Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky solve · reused factor

128 × 128, 1 RHS · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.999471067386015	0.9973565952947396 – 1.0020628371945415

Cholesky solve · reused factor

256 × 256, 64 RHS · f64

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.083568245363017	1.0802557742871344 – 1.08517793042905

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

LU solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.996840299031319	0.9959151607706278 – 0.9975877328672669

Thin QR

512 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9967563158814609	0.9918963979065318 – 1.0046826330122811

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Thin QR

1024 × 256 · f64

Observed time reduction

CURRENT / BASELINE	**0.9871204439911031**	95% INTERVAL	0.9779744047084407 – 0.9891396504940139

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	**0.9998129801151108**	95% INTERVAL	0.9957627512381872 – 1.0028547266517467

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0028165236051503	0.9960981496379726 – 1.0098207241763921

Thin QR

256 × 64 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9987499230413007	0.9846317345901487 – 1.0057244869354816

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

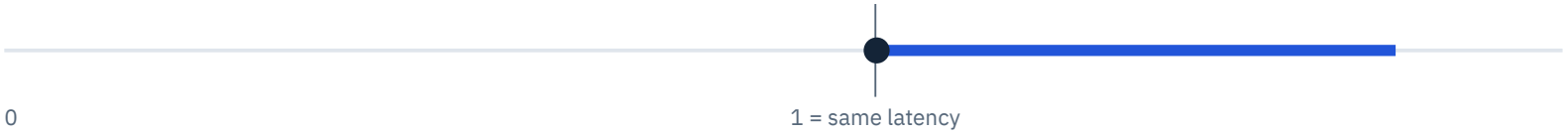
Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9963023829087921	0.9837728194726166 – 1.0041597337770383

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0013946850606525	0.9919242970052153 – 1.597163381838276

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.000878445152081	0.9969261170271161 – 1.0062039122179538

Matrix multiplication

256 × 256 · 256 × 256 · f64

Observed time reduction

CURRENT / BASELINE	95% INTERVAL
0.9981894802135081	0.9808812825940344 – 0.9992079677625157

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0007649743733584	0.996856598178352 – 1.001591439699754

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9972580202906498	0.9864755206924534 – 1.0137098985467508

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0011809544258952	0.994669793721017 – 1.0053511348464619

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.000462300495322	0.9910861814070788 – 1.0065968862696808

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0049407114624507	0.9856648541769648 – 1.009930486593843

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9916805324459235	0.9916107382550335 – 1.007607776838546

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.000831230693046	0.9978279884237653 – 1.004733574143589

Cholesky factorization

256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0035265993610598	0.9962365833107335 – 1.0075267678861823

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Observed time increase

CURRENT / BASELINE	95% INTERVAL
1.009350517607689	1.0043837900899306 – 1.0151540833691663

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
0.9973336177474402	0.9863506001487095 – 1.0044559308439533

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.00182105104314	0.9877555854589073 – 1.0068579077147009

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear

CURRENT / BASELINE	95% INTERVAL
1.0022567541427556	0.9981015509363537 – 1.004204377687913